

MAFE Financial Econometrics

Maximum Likelihood Estimation (MLE)

Ghasan Asbool

University of Oxford

10/06/2026



MLE - Learning Road Map

1. Lecture:

- An introduction to MLE with a Poisson distribution example.

2. Pre-class take-home exercise:

- Solidify your understanding of the lecture material in preparation for the applied class exercise.

3. Applied class exercise:

- Estimation of market return and risk from the S&P 500 data.



Teaching

- Structured rigour: Question $\rightarrow$ method $\rightarrow$ implementation $\rightarrow$ interpretation.
- Teaching material that is not typically within my domain.
- Inclusive access without dilution.
- Engagement and interactivity as key elements of teaching delivery.
- Supervision: structured thinking and mapping concepts to final implementation.



Administration

- Not merely clerical compliance tasks.
- Examinations and fairness of the assessment.
- Admissions: good selection, but also fair treatment.
- Mechanisms for supporting the teaching of colleagues.
- Pastoral duties and targeted support.



Academic citizenship

- Ongoing research with policy relevance.
- Connecting Teaching and Research.
- Identifying unconventional funding and revenue sources.
- Connecting with the community and enriching discussions on current world issues.



Thank you!

